

About electrical rockets (ER)

1- some electric rockets such as arc-jet increases T_0 , this increases u_e as $u_e = \sqrt{2c_p(T_0 - T_e)}$, but they usually do not increase P_0 (like chemical rockets) &

$\dot{m} \propto \frac{P_0}{\sqrt{T_0}}$ (choked nozzle), so $\dot{m}$ does not increase

in ER's, and Z does not increase!

But I_{sp} goes up $\uparrow \rightarrow I_{sp} = \frac{u_e}{g}$

we can write $I_{sp} = \frac{\dot{m} u_e}{\dot{m} g_e} = \frac{Z}{\dot{m} g_e}$ as $\dot{m}$ is small

I_{sp} is large!

2 most important relation for ER:

$$\eta = \frac{kE)_e}{P} < 1$$

How much power supplied is converted to $kE)_{exit}$

$$\alpha = \frac{M_p + M_s}{P} > 1$$

what is the ratio of mass that creates the power is compared to the power!

we want $\eta \uparrow \Rightarrow \alpha \downarrow$

$$M_L = M_o - M_p$$

$$M_b = M_L + (M_p + M_s) = 139.3 \text{ kg}$$

$$\frac{M_L}{M_o} = 0.61$$

$$\Delta u_b = u_e \ln \frac{M_o}{M_b} = 19620 \ln \frac{164}{139.3} = 3203 \frac{\text{m}}{\text{s}} = 3.2 \frac{\text{km}}{\text{s}}$$

$$\text{Table 10.8} \Rightarrow (\Delta u_b)_{\text{req}} = 4.2 \text{ km/s}$$

No, the vehicle cannot raise the payload from LEO to GEO.



$$\eta = 0.5 = \frac{k.E)_e}{P}$$

half the power is converted to $k.E)_e$

$$\alpha = 10 \frac{\text{kg}}{\text{kw}} = \frac{(M_p + M_s)}{P}$$

means for 1 kw power, we add 10 kg mass